Pratik Palwai

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Github: github.com/Pratik-Palwai/ | CAD Projects: grabcad.com/pratik.palwai-1/models

Education

University of California, Irvine (Graduating June 2028)
Mechanical Engineering, B.S. | Minor in Information/Computer Science | GPA 3.7

Experience

Off-Road Engineering Intern, Hyundai Motor Group *(June 2026 - Present)*
- Developed and tuned traction control software for the 2028 Hyundai Santa Fe XRT terrain modes, modulating brake pressure and powertrain torque to maintain vehicle stability and controlled wheel spin
- Instrumented a 2025 Ford Bronco Badlands with 40+ strain gauges, developing a calibration procedure for driveshaft torque measurements and collecting empirical load data for target setting
- Developed a Python library for automated CANbus data acquisition, sensor signal filtering, and rapid test reporting

Aircraft Lead, UCI UAV Forge SUAS *(April 2026 - Present)*
- Led all technical aspects of the aircraft for each year's challenge, including sizing optimization, subsystem design and prototyping, and flight verifications
- Applied engineering first principles in Python, Flow5, and MIT xFoil to size wing and tail geometries, optimizing cruise and stall conditions while satisfying structural load and payload constraints

Machine Shop Staff, UCI Samueli School of Engineering *(Jan 2025 - Present)*
- Mentored 200+ students on hardware design, 3D CAD modeling, and fabrication of pneumatically propelled robots
- Instructed students on CNC mill and 3D printer operation, verifying dimensional tolerances using calipers, micrometers, and dial indicators
- Performed preventative maintenance, failure analysis, and repairs to ensure shop machinery was always in safe, working order

Teaching Assistant, UCI EECS Department *(Sept 2025 - Present)*
- Guided students in electromechanical assembly, firmware configuration, and flight testing autonomous drones
- Evaluated onboard AI vision integration with OpenCV on Raspberry Pi for real-time object tracking
- Mentored students in debugging hardware, firmware, and peripheral communication over UART and I2C interfaces

Chief Engineer, UCI AIAA Design/Build/Fly *(Sept 2024 - April 2026)*
- Led all technical aspects of the aircraft for each year's challenge, including sizing optimization, subsystem design and prototyping, and flight verifications
- Designed and built a custom 36" 4-axis CNC router with RP2040 microcontroller and custom grblHAL firmware to precision-cut foam cores, reducing composite wing structural mass by 40%

Powertrain Lead, UCI Formula SAE EV Racing *(Sept 2024 - April 2026)*
- Generated 2D drawings using GD&T per ASME Y14.5, machined powertrain hardware, and designed a go/no-go fixture to inspect machined parts for proper fit and tolerances
- Simulated vehicle acceleration in Python to optimize gear ratios, enabling 0–60 mph launches in under 2.50 sec
- Implemented closed-loop PID control for coolant pumps and radiator fans to regulate motor and inverter thermal limits while minimizing electrical power draw

Awards: Eagle Scout (November 2023), ROP Engineering Student of the Year (May 2024)

Skills

CAD/CAE: 2D Drawings & 3D CAD (SolidWorks, CSWP), GD&T, FEA/CFD, MATLAB, XFLR5
Fabrication/Metrology: CNC Machining, Manual Lathes/Mills, Precision Metrology, Composites, 3D Printing
Software/Hardware: Python, C++, CANbus, Microcontrollers (RP2040, ESP32, RPi), Oscilloscopes, PID Control